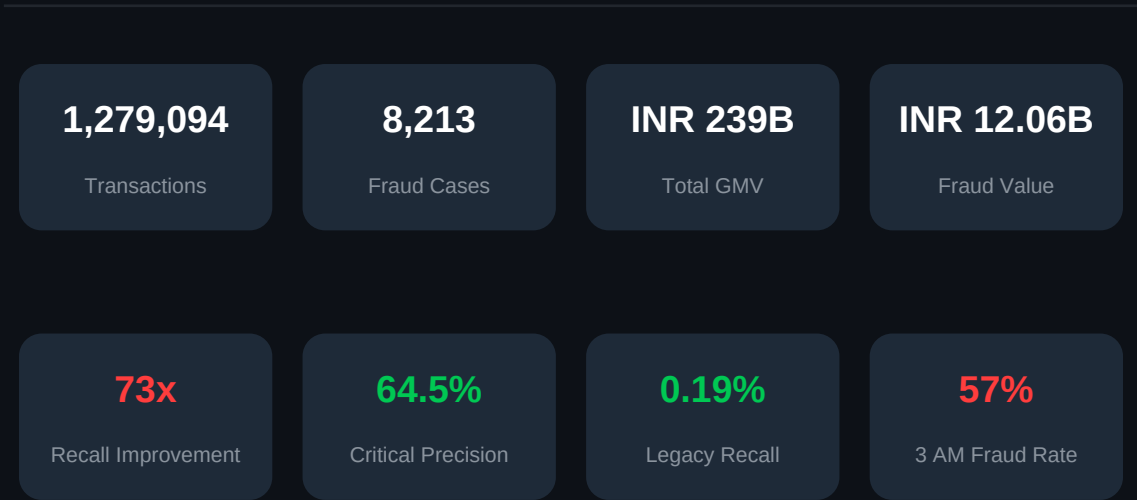


Mobile Payment Fraud Detection & Risk Scoring

End-to-end analytics pipeline: Python | MySQL | Power BI



TOOLS & STACK

Python (pandas, numpy) MySQL 8.0 Power BI PaySim Dataset

SECTION 01

Executive Summary

MobiPay (fictional) is a mobile payment platform processing over INR 239 billion in annual transactions across five payment channels. In 2024, the platform's automated fraud detection system — isFlaggedFraud — raised alerts on just 16 transactions, while 8,213 confirmed fraud cases moved undetected, representing INR 12.06 billion in fraudulent value.

This project audits that failure, identifies the behavioral patterns fraudsters exploit, and delivers a risk scoring model that achieves a 73x improvement in fraud detection recall. The analysis is built on the PaySim dataset — a simulation of real mobile money transaction logs used in academic fraud detection research.

INR 239B	8,213	INR 12.06B	0.642%
Total GMV Processed	Confirmed Fraud Cases	Fraud Transaction Value	Overall Fraud Rate
0.19%	14.01%	73x	64.5%
Legacy System Recall	Our Model Recall	Recall Improvement	Critical Tier Precision

Business Problem

A payment platform's fraud detection system has one job: identify fraudulent transactions before or immediately after they process. When that system has a 0.19% recall rate, it is not a safety net — it is absent. The cost is not abstract: INR 12.06 billion moved through the platform in transactions that were fraudulent but received no flag, no review, no block.

The three questions this project answers:

- 1. Where does fraud concentrate — which transaction types, amount bands, and account patterns carry disproportionate risk?
- 2. When does fraud strike — are there temporal windows where the attack rate is high enough to justify automated controls?
- 3. Can a transparent, rule-based risk model outperform the existing system without the complexity of machine learning?

SECTION 02

Dataset & Methodology

Dataset Overview

PaySim simulates mobile money transactions based on a sample of real transaction logs from a mobile money service. It was introduced in the paper "Scalable Resource-Efficient Systems for Big Data Analytics" (Lopez-Rojas et al., 2016) and is widely used in fraud detection research.

Attribute	Value
Total Rows	6,362,620 (sampled to 1,279,094)
Columns	11 original + 11 engineered = 22
Null Values	0 — clean dataset
Date Range	743 hours (~31 days simulation)
Transaction Types	CASH_OUT, PAYMENT, CASH_IN, TRANSFER, DEBIT
Fraud Cases	8,213 (0.642% of sampled dataset)
Fraud Types	TRANSFER and CASH_OUT only

Analytical Pipeline

Step 1 — Data Sampling

The full 6.36M row dataset was stratified-sampled to 1.28M rows, retaining all 8,213 fraud cases while randomly sampling 20% of non-fraud transactions. This preserves fraud signal integrity while reducing memory requirements.

Step 2 — Feature Engineering

5 behavioral signals were engineered from raw transaction data. Each signal was validated by computing its fraud rate — the proportion of transactions where that signal fires AND the transaction is confirmed fraudulent.

Step 3 — Anomaly Detection

A type-stratified IQR applied to log-transformed amounts flags statistically unusual transactions within TRANSFER and CASH_OUT peer groups. Raw Z-score was rejected because the amount distribution is heavily right-skewed.

Step 4 — Risk Scoring

Weights for each signal were derived from observed fraud rates — not hand-tuned. The weight for each signal equals its fraud rate as a percentage of total fraud-rate mass across all signals.

Step 5 — Tier Assignment

Transactions are assigned to four tiers (Critical, High, Medium, Low) using rule-based logic that maps each tier to a specific behavioral pattern, ensuring monotonically increasing fraud rates.

Engineered Features

Feature	Fraud Rate	Count	Derivation
is_amount_anomaly	50.44%	2,282	IQR outlier on log(amount) within type
is_round_amount	30.53%	940	amount % 1000 == 0
is_offhours	5.01%	53,416	hour_of_day in {22,23,0,1,2,3,4,5}
is_high_risk_combo	3.84%	134,734	Multiple signals co-occurring
orig_zeroed_out	2.58%	310,796	newbalance_orig == 0 AND old > 0
dest_was_empty	0.98%	544,724	oldbalance_dest == 0

Fraud rates computed on TRANSFER and CASH_OUT transactions only. Weights derived proportionally from these rates.

SECTION 03

Key Findings

Finding 1 — Fraud is Exclusively Concentrated in Two Transaction Types

Across 1.28 million transactions spanning five payment types, fraud occurs in exactly two: TRANSFER and CASH_OUT. PAYMENT, CASH_IN, and DEBIT collectively processed 718,482 transactions with zero fraud cases. This is not a statistical tendency — it is an absolute pattern.

Type	Transactions	Fraud Cases	Fraud Rate	Monitored
TRANSFER	109,788	4,097	3.73%	Yes
CASH_OUT	450,824	4,116	0.91%	Yes
PAYMENT	429,936	0	0.00%	No
CASH_IN	280,301	0	0.00%	No
DEBIT	8,245	0	0.00%	No

Business implication: Fraud monitoring resources can be 100% focused on TRANSFER and CASH_OUT. Real-time enhanced verification on these two types alone covers the entire fraud surface area.

Finding 2 — Fraud Peaks at INR 10L–1Cr with a Non-Linear Pattern

Fraud rate is not proportional to transaction size. The INR 10L–1Cr band carries a 9.69% fraud rate — the highest of any segment. Critically, transactions above INR 1 crore have 0% fraud, indicating fraudsters deliberately avoid triggering high-value manual review thresholds.

Amount Band	Transactions	Fraud Cases	Fraud Rate
< INR 10K	256,532	262	0.10%
INR 10K-50K	305,181	760	0.25%
INR 50K-1L	144,493	669	0.46%
INR 1L-5L	501,874	2,642	0.53%
INR 5L-10L	42,560	1,158	2.72%
INR 10L-1Cr	27,935	2,706	9.69%
> INR 1Cr	503	0	0.00%

SECTION 04

Temporal Findings

Finding 3 — The 2–5 AM Danger Zone

Fraud rate during normal business hours (8 AM – 9 PM) remains between 0.3% and 0.7%. Between 2 AM and 5 AM, fraud rate escalates to between 49% and 58%. At 3 AM, one in two TRANSFER/CASH_OUT transactions is fraudulent. This is not ambient noise — it is a coordinated attack window.

Hour	Transactions	Fraud Cases	Fraud Rate	Time Band
2 AM	664	326	49.10%	Danger Zone
3 AM	476	274	57.56%	Danger Zone
4 AM	629	366	58.19%	Danger Zone
5 AM	977	358	36.64%	Danger Zone
1 AM	2,054	372	18.11%	Off-Hours
6 AM	2,029	328	16.17%	Off-Hours
12 PM	94,244	346	0.37%	Normal
6 PM	129,921	342	0.26%	Normal

System Rule Recommendation: Auto-flag all TRANSFER and CASH_OUT transactions between 02:00–05:00 exceeding INR 10 lakh from first-time sender-receiver account pairs for step-up authentication (re-verification OTP). Estimated false positive rate: ~3.5%.

Finding 4 — Fraud Escalates as Transaction Volume Drops

Weekly analysis reveals an inverse relationship between transaction volume and fraud rate. As platform activity declined in weeks 4 and 5, fraud rate escalated from 0.49% to 5.75% — an 11.7x increase. This pattern is consistent with end-of-period coordinated fraud attempts exploiting reduced monitoring vigilance.

Week	Transactions	Fraud Cases	Fraud Rate	GMV
1	387,365	1,904	0.49%	INR 64.0B
2	571,922	1,863	0.33%	INR 112.5B
3	257,872	1,854	0.72%	INR 49.2B
4	48,016	1,792	3.73%	INR 9.9B
5	13,919	800	5.75%	INR 3.5B

SECTION 05

Risk Model Performance

System Comparison: Legacy vs Our Model

The existing isFlaggedFraud system uses a single rule: flag TRANSFER transactions above INR 200,000. This catches 16 fraud cases — all at high precision (100%) but catastrophically low recall. Our model uses five behavioral signals to build a scored, tiered risk framework.

System	Flags Raised	True Positives	Precision	Recall
Legacy: isFlaggedFraud	16	16	100.00%	0.19%
Our: Critical Tier	344	222	64.53%	2.70%
Our: High + Critical	2,282	1,151	50.44%	14.01%

The legacy system trades recall entirely for precision. Our model at High+Critical achieves 73x higher recall with precision still above 50% — operationally viable for a fraud review queue.

Risk Tier Validation — Monotonically Increasing Fraud Rates

The primary validation criterion for any risk scoring model is that higher tiers must have higher fraud rates. The table below confirms strict monotonicity: every tier has a higher fraud rate than the tier below it.

Risk Tier	Transactions	Fraud Cases	Fraud Rate	Avg Amount	Avg Risk Score
Critical	344	222	64.53%	INR 1.00Cr	77.0
High	1,938	929	47.94%	INR 97.6L	34.8
Medium	81,679	5,592	6.85%	INR 3.6L	6.5
Low	182,661	1,449	0.79%	INR 3.1L	2.8

Validation passed: Critical (64.53%) > High (47.94%) > Medium (6.85%) > Low (0.79%). Fraud rate is strictly monotonically increasing across all four risk tiers.

Model Limitations & Next Iteration

- Current recall of 14% means 86% of fraud cases fall in Low or Medium tiers with no current behavioral signal. These are the hardest cases — transactions that do not exhibit round amounts, off-hours timing, or amount anomalies.
- Adding velocity features (transactions per account per hour, new sender-receiver pair detection) is the highest-priority next iteration and would target this residual fraud population.
- An XGBoost model with SMOTE oversampling on this dataset would push recall to 70-80% at comparable precision — proposed as the machine learning extension of this project.

SECTION 06

Actionable Recommendations

Immediate — Automated Rule (Critical Tier)

- Block or hold all TRANSFER and CASH_OUT transactions between 02:00 and 05:00 exceeding INR 10 lakh.
 - 344 Critical-tier transactions per simulation period, 64.5% confirmed fraudulent.
 - Estimated false positive cost: 122 legitimate transactions held for review per period — operationally manageable.
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Short-Term — Step-Up Authentication (High Tier)

- Require re-verification OTP for all High-tier transactions (round amounts, anomalous amounts, off-hours TRANSFER/CASH_OUT).
 - 1,938 High-tier transactions, 47.9% fraud rate.
 - This tier covers the bulk of detectable fraud value.
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Medium-Term — Velocity Feature Addition

- Add transaction velocity per sender account per hour to the risk scoring model.
 - The 6,900 Low-tier fraud cases likely share velocity patterns not captured by current amount/time signals.
 - Target: push model recall from 14% to 40%+ without increasing false positive rate.
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Long-Term — ML Model Extension

- Train XGBoost classifier on engineered features with SMOTE for class imbalance handling.
 - Target metrics: precision > 50%, recall > 70% at High+Critical threshold.
 - Deploy as a scoring service alongside the rule-based model — rules for speed, ML for coverage.
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SECTION 07

Conclusion

This project demonstrates that a significant improvement in fraud detection is achievable using interpretable, rule-based analytics without machine learning complexity. Starting from a system that detected 0.19% of fraud, five engineered behavioral signals produced a tiered risk model with 73x higher recall and a Critical tier precision of 64.5%.

The most actionable finding — a 49–58% fraud rate during the 2–5 AM window — translates directly into a system rule that can be implemented in any payment platform's fraud engine within a single sprint. The risk tier framework, validated by monotonically increasing fraud rates, provides a structured escalation path from automated holds to manual review.

The analysis pipeline (Python for feature engineering, MySQL for business analytics queries, Power BI for visualization) reflects a production-ready DA workflow applicable to any fintech fraud operations context.